

Input Ontology (IO)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: ARABICMMLU | **Context:** MENA Multi-Country Arabic Tutoring System — Eight-Country Curriculum Deployment

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

ArabicMMLU's taxonomy of 40 tasks across STEM, Social Science, Humanities, Arabic Language, and Other [Q25, Q107], spanning primary through university levels, broadly matches the deployment's educational tiers and subject scope. All six university subjects required by the deployment (law, CS, management, economics, political science, accounting) are confirmed present in the benchmark configurations [DATASET-D8, D11, D13, D14, D16]. However, university-level questions constitute only ~6.1% of the benchmark [Q71], yielding roughly ~889 questions spread across all university subjects — thin coverage for a deployment that explicitly targets university students. Country coverage at the ontology level is structurally uneven: Jordan dominates with over 6,000 questions while some countries contributed 'as few as 100 or, in some cases, none at all' [Q89]. Several configurations have null country/level metadata [DATASET-D26, D27], limiting their use for country-specific diagnostics. The taxonomy includes no madhab labeling for Islamic studies and no jurisdiction-specific branching for law, both flagged as HIGH-priority deployment requirements.

ID	Evidence
Q25	"ArabicMMLU is an Arabic multiple-choice question-answering dataset comprising 40 tasks spanning a wide range of subjects and education levels." (p.3)
Q71	"This could be attributed to the relatively small portion (i.e., 6%) of university-level questions in ArabicMMU, which potentially skews the results." (p.7)
Q89	"ArabicMMLU does not represent all Arabic countries equally. For example, we have collected over 6K multiple-choice questions from Jordan, while other countries are represented with only 100 questions or, in some cases, not at all." (p.9)
Q107	"Table 9 presents results organized by subject groups: STEM, Social Science, Humanities, Arabic Language, and Other." (p.16)
D8	— Moroccan procedural law — appeals rules under Moroccan Code of Civil Procedure يؤدى لدل رهوج يف مكحل رودص دعب الإ عوفد وأ ضراع عازن يف ةرداصل وأ ةيدي مهمم التل ماكحلأ فانئتسا لبقي ال
D11	— "Sequential events, cooperative or conflictual, are called: interactions" — general IR concept تالعات التل: هى ةى عارض وأ ةى نواع نوكت دق ةع باتم تادحا
D13	— Computer auditing concept — general accounting principle, from Egyptian source ةى عبت التل اب ةمى مس نوكت لي غشتل ةى ممع نإف... ةمى مس تل اخدم تل ناك اذا ونأ بساحل لوح ةع جارمل بومسأ ضر تفي
D14	— Severely OCR-distorted text about authority distribution in organizations, from Egyptian source لى إ ري تذت... ةتة تل خسل ةت ي رادلإ ةا ي هتة ت دسل او صاخ تل ألأ ني تب ةطل تل ةت ي هت ةت جرد
D16	— "A microcomputer is a desktop computer" — from KSA source (faculty.ksu.edu.sa) التل كمكالم رتوي بكمكل زاهج: ن ع ةرابع وه قيق دل رتوي بكمكل
D26	All 5 examples: Country = null, Source = folderat.com, Subject = Islamic Studies — Islamic Studies "All" config examples have no country metadata — source appears to be generic Arabic quiz site
D27	All 5 examples: Country = null, Source = madinaharabic.com — Arabic Language General has no country metadata — sourced from online Arabic language learning site

Strengths

- All six deployment-required university subjects are present as queryable configurations [DATASET-D8, D11, D13, D14, D16]

- Multi-level coverage (primary, middle, high school, university) is empirically confirmed in the data [DATASET-D5, D13, D21]
- Subject taxonomy explicitly includes country-variation-prone subjects: history, geography, civics, political science [Q72, Q73]
- Negation is treated as a distinct linguistic phenomenon [Q83, Q84], providing some construct depth beyond factual recall

ID	Evidence
Q72	"We present the performance of open-source models on selected subjects that potentially contain Arabic-specific contexts." (p.7)
Q73	"These subjects include history, geography, civics, political" (p.7)
Q83	"Despite negation being an absolutely foundational linguistic phenomenon, LLMs have been shown to be worryingly insensitive to its effects in English (Kassner and Schütze, 2020; Hosseini et al., 2021; Truong et al., 2023)." (p.8)
Q84	"We thus perform an analysis of LLM performance over questions in ArabicMMLU with and without negation to determine whether this observation ports across to Arabic." (p.8)
D5	"كلمة الملك عبد الله الأول ملك الأردن في 1952 مع روتس د رص" with options referring to Jordanian kings — "The 1952 constitution was issued during the reign of King [Talal bin Abdullah]" — Jordanian royal history
D8	— يوعدل رهوج في مكحل رودص دعب الإ عوفد وأ ضراع عازن في أرداصلا وأ ديديهمم التل ماكحلأ فانئتسا لبقي ال Moroccan procedural law — appeals rules under Moroccan Code of Civil Procedure
D11	— "Sequential events, cooperative or conflictual, are called: interactions" — general IR concept
D13	— عيبتل اب قمي مس نوكت لي غشتل ايمم نإف... قمي مس تل اخدم تل تنك اذ ونأ بساحل ل و ح ع ج ارم ال بوم س أ ضرت في Computer auditing concept — general accounting principle, from Egyptian source
D14	— ل ي ل ي ت ذ ت... ق ت ف ل ت خ س ل ل ق ت ي ر ا د ل إ ل ت ا ي ه ت د س ل ل و ص ا خ ل ت أ ل أ ن ي ت ب ط ل ت د ل ع ت ي ي ه ت ق ت ج ر د OCR-distorted text about authority distribution in organizations, from Egyptian source
D16	— ي ب ك م ال ر ت و ي ب ك م ال ز ا ه ج : ن ع ق ر ا ب ع و ه ق ي ق د ل ر ت و ي ب ك م ال "A microcomputer is a desktop computer" — from KSA source (faculty.ksu.edu.sa)
D21	— ق ن س ه م ك ح د م و م 1952: في كل ذوقية حص باب سأل هتيالو تهنتا نيسجل نبال ط ل ل م ل "King Talal bin Hussein's reign ended for health reasons in 1952 — one year" — Jordanian royal history in primary school

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories include MSA-medium primary/middle/high school content for Morocco, Egypt, Jordan, Palestine, Lebanon, UAE, Kuwait, KSA across STEM, humanities, civics, Islamic studies, Arabic language; plus university-level coverage for law, CS, management, economics, political science, accounting [elicitation A1, A2].

IO-2: Several regionally-relevant categories are omitted or thinly covered: madhab-specific Islamic studies tracks (Maliki for Morocco, Hanbali for KSA) have no labeling [gap_id 7]; jurisdiction-specific law branches are absent (flat label schema [Q27]); Kuwaiti civic/national content appears absent — no Kuwaiti workers documented [Q31, Q96] and no Kuwaiti examples observed in data sample [DATASET content summary]; French-medium Moroccan university science content is fully out of scope [Q92].

IO-3: Some included categories are not directly relevant to all deployment cohorts: 'Driving Test' configuration [DATASET-D35] is not a tutoring subject; null-country generic Islamic Studies and Arabic Language items [DATASET-D26, D27] cannot be used for country-stratified evaluation. These are not harmful inclusions per se but contribute construct-irrelevant variance to

aggregate scores.

IO-4: Documented gaps: (a) university-level coverage at ~6.1% [Q71] is structurally thin for a deployment serving university students; (b) no madhab calibration [gap_id 7]; (c) no jurisdiction-specific law taxonomy [Q27, gap_id 4]; (d) Kuwait coverage likely near-zero [Q89, Q96]; (e) French-medium Moroccan content fully absent [Q92].

ID	Evidence
Q25	"ArabicMMLU is an Arabic multiple-choice question-answering dataset comprising 40 tasks spanning a wide range of subjects and education levels." (p.3)
Q27	"Each question has 2–5 candidate answers, with one correct answer." (p.3)
Q31	"The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)
Q71	"This could be attributed to the relatively small portion (i.e., 6%) of university-level questions in ArabicMMU, which potentially skews the results." (p.7)
Q89	"ArabicMMLU does not represent all Arabic countries equally. For example, we have collected over 6K multiple-choice questions from Jordan, while other countries are represented with only 100 questions or, in some cases, not at all." (p.9)
Q92	"The dataset primarily focuses on Modern Standard Arabic (MSA). However, multilingual and Arabic LLMs are often exposed to both MSA and dialectal Arabic." (p.9)
Q96	"We extend our gratitude to all collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia who participated in the data collection process." (p.10)
Q107	"Table 9 presents results organized by subject groups: STEM, Social Science, Humanities, Arabic Language, and Other." (p.16)
D26	All 5 examples: Country = null, Source = folderat.com, Subject = Islamic Studies — Islamic Studies "All" config examples have no country metadata — source appears to be generic Arabic quiz site
D27	All 5 examples: Country = null, Source = madinaharabic.com — Arabic Language General has no country metadata — sourced from online Arabic language learning site
D35	رتم 50 دعب ىلع؟ ععضو كىل ع نيأ، سكاغلا ريذحتلا ثلثم كىدلو كتبكرم لاطعت لاج يف — UAE driving rule (50m warning triangle placement) — country-specific traffic regulation
WEB-1	Models perform notably poorly on Morocco- and UAE-sourced questions, consistent with thin annotation coverage (https://arxiv.org/html/2402.12840v1)
WEB-4	Arabic LLM benchmark survey confirms no French-Arabic bilingual benchmark for Moroccan higher education exists (https://arxiv.org/abs/2510.13430)

Information Gaps

- Exact per-country question counts beyond Jordan (>6,000) are not text-published [gap_id 5]
- Madhab calibration of Islamic studies questions is undocumented anywhere [gap_id 7]
- Jurisdictional distribution of law questions beyond Moroccan procedural law sample is unknown

Requires Expert Verification

- Whether observed Moroccan-only Law (Professional) sample generalizes to the full ~299 law questions or whether other configurations contain embedded jurisdiction-specific content
- Whether Islamic Studies configurations contain madhab-specific fiqh that would diverge across Maliki/Hanbali/Shafi'i traditions

Remediation

Gap 1: University-level coverage is ~6.1% of total [Q71] (~889 questions across all subjects), and Law (Professional) is exclusively Moroccan in the sample [DATASET-D34]; jurisdiction-specific tagging is absent.

Recommendation: Supplement with ArabLegalEval [WEB-3] for Saudi law, MizanQA [WEB-4] for Moroccan law, and AraSTEM [WEB-2] for STEM university content. Commission custom evaluation sets for Egyptian, UAE, Jordanian, Lebanese, Kuwaiti, and Palestinian legal jurisdictions which remain entirely unfilled by existing Arabic benchmarks.

ID	Evidence
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Q71	"This could be attributed to the relatively small portion (i.e., 6%) of university-level questions in ArabicMMU, which potentially skews the results." (p.7)
D34	All 5 Law (Professional) examples: Country = Morocco, from single Google Drive PDF — All sampled law questions are exclusively from Morocco — single-source PDF
WEB-2	https://arxiv.org/abs/2501.00559
WEB-3	ArabLegalEval reuses ArabicMMLU law subsets without jurisdiction tagging — confirms multi-jurisdiction gap is structural (https://arxiv.org/abs/2408.07983)
WEB-4	Arabic LLM benchmark survey confirms no French-Arabic bilingual benchmark for Moroccan higher education exists (https://arxiv.org/abs/2510.13430)